

Table 1: Protein homology accuracy on BioPAWS [wang2026biopaws], comparing classical alignment, supervised PLMs, MoE bio-models, and OmniGene-4 stages. *Standard* uses `protein_pair_short` (close homologs); *Remote* uses `protein_pair_remote` (structurally related but sequence-divergent). “Compute” is order-of-magnitude GPU-days for the listed checkpoint; for classical tools it is search-time on a single CPU run. Bold = best in column; *italic* = our work.

Method	Type	Standard (%)	Remote (%)	Compute (GPU-days)
<i>Classical alignment (no learning)</i>				
MMseqs2 [steinegger2017mmseqs2]	alignment	—	54.40	—
DIAMOND [buchfink2021diamond]	alignment	—	53.20	—
<i>Pre-trained protein language models</i>				
ESM-2 650M [lin2023evolutionary]	dense PLM	—	50.50	$\sim 10^3$
ESM-2 3B [lin2023evolutionary]	dense PLM	—	51.20	$\sim 10^4$
<i>General LLMs (literature-only baselines)</i>				
GPT-2 + PAWS-X SFT [wang2026biopaws]	dense LM	84.00	—	$\sim 10^0$
LLaMA-3.1 8B + LoRA [wang2026biopaws]	dense LM	72.00	—	$\sim 10^0$
Gemma-4-26B-Instruct (vocab-extended, zero-shot)	MoE LM	85.00	60.00	0
Qwen-3 (massive)	MoE LM	~ 100	~ 75	$\sim 10^4$
<i>Specialized MoE bio-foundation models</i>				
AIDO.Protein-16B [sun2024aido]	MoE PLM	—	— [†]	$\sim 1.6 \times 10^4$
<i>OmniGene-4 stages (this work, single H20 GPU)</i>				
OmniGene-4 v2 (CPT + Bio-SFT)	MoE bio	100.00	56.55	0.5
OmniGene-4 v3 (+20K remote)	MoE bio	99.95	59.50	0.6
OmniGene-4 v4 (Alpaca + loss-mask)	MoE bio	99.50	82.00	1.3
OmniGene-4 v5 (dual-head)	MoE bio	99.40	82.60	1.5
<i>OmniGene-4-MM stages (this work, multi-modal extension)</i>				
MM Stage 1 (vision warmup)	MoE bio+V	— [‡]	— [‡]	0.4
MM Stage 2 (mixed text + vision)	MoE bio+V	59.00	56.50	1.0
MM Stage 3 v2 (LR 5e-6, embed trainable)	MoE bio+V	59.50	60.00	2.3
MM Stage 3 v3 (LR 2e-5, embed frozen)	MoE bio+V	85.00	69.50	1.9

[†] AIDO.Protein [sun2024aido] does not report on the BioPAWS homology task; results listed in their paper cover xTrimoPGLM and ProteinGym DMS benchmarks. The compute figure ($256 \text{ A100} \times 64 \text{ d} \approx 16,384 \text{ GPU-days}$) reflects pre-training cost only.

[‡] Stage 1 evaluates vision capabilities only; the homology head is not yet recovered from vision-warmup catastrophic forgetting (4/200 valid text generations).

Table 2: Multi-modal capability across OmniGene-4-MM training stages. Vision metrics are accuracy on a 50-prompt-per-task subset of Vis-CheBI20 [fan2025ocsu]. Multi-task generation is keyword-overlap score on a 30-prompt subset of an internal SFT-eval set covering biological domains. “Stage 3 v3” is the final model recommended for downstream use.

Stage	Vision (Vis-CheBI20)					Multi-task generation (kw-score)				
	struct recog	struct cap	general desp	trans IUPAC	trans SMILES	Cell	Mol	Protein (gen)	Lit. (gen)	Hom.
v5 (text only)	—	—	—	—	—	—	—	—	—	99.4 / 82.6
MM Stage 1 (vision)	1.00	0.90	0.10	0.00	0.00	—	—	—	—	—
MM Stage 2 (mixed)	1.00	0.88	0.14	0.00	0.00	0.25	0.32	1.00	0.17	59.0 / 56.5
MM Stage 3 v2	1.00	0.90	0.16	0.00	0.00	0.75	0.79	1.00	0.24	59.5 / 60.0
MM Stage 3 v3	1.00	0.96	0.14	0.00	0.00	0.95	0.91	1.00	0.30	85.0 / 69.5

The “trans_IUPAC” and “trans_SMILES” columns measure exact-match accuracy of generated chemical strings; near-zero values reflect a known limitation of LoRA-tuned LLMs on character-level chemical naming and motivate the OCSU [fan2025ocsu] reframing in the discussion. “Hom.” shows BioPAWS standard / remote accuracy carried over from Table ??.

Table 3: Methodological comparison of recent Mixture-of-Experts bio-foundation models. “Modality” lists the input types the model can natively process. “Mechanism” asks whether the paper’s central claim is about a transferable *mechanism* or about a downstream *tool*. Compute is for the listed checkpoint, including pre-training when applicable; we use one significant figure.

	AIDO.Protein [sun2024aido]	Tripathi et al. [tripathi2025tfbs]	OmniGene-4 / MM (this work)
Year	2024	2025	2025
Modality	Protein only	DNA only	8 modalities (vision, sequence, language)
Backbone	16B-A8B sparse MoE (from scratch)	MoE of CNNs (DeepBIND ensemble)	Gemma-4 26B-A4B MoE + LoRA r=64
Pre-train data	1.2T tokens	—	96 GB mixed (DNA / protein / OWT / 3Di / DSSP); ~100M-token paraphrase fine-tune
Compute	$\sim 1.6 \times 10^4$ GPU-days	$\sim 10^1$ GPU-days	~ 1.5 GPU-days
Goal	SOTA on PEER, ProteinGym DMS	TFBS classification stability	Modality-invariant transfer + tool
Transfer claim	no	no	yes (paraphrase $\rightarrow$ homology, robust under modality scaling)
Multi-modal	no	no	yes (4 vision + 3 sequence + 1 NL)
Interpretability	—	ShiftSmooth (gradient attribution)	Router-level expert decomposition (3-tier modality emergence)
Code / weights public	yes (HF, GitHub)	—	yes (HF, GitHub)
Mechanism vs tool	tool	tool	mechanism + tool